# Firm Wage Setting and On-the-Job Search Limit Wage-Price Spirals: An Analytic Two-Period Model (Not for Publication)

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**Abstract**

We provide a simplified two-period version of Bloesch, Lee and Weber (2025) and analytically prove the main propositions.

## 1 Two-Period Model

In this section, we build a simple two-period general equilibrium model that illustrates the following two features of Bloesch, Lee and Weber (2025) in a sharper way:

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1. When employed and unemployed workers share consumption risks such that $\frac{C_t^e}{C_t^u} = \xi$—meaning the unemployed receive a fraction $\xi^{-1}$ of the consumption of employed workers—the cost-of-living shock generally has no effect on wages or labor market outcomes.

   — The baseline result in Bloesch et al. (2025).

2. When the unemployment benefit $b_t$ is expressed in real terms and compared by workers with the real wage $\frac{W_t}{P_t}$ when workers decide whether to join the workforce, a cost-of-living shock triggers a positive wage response. However, this wage pass-through diminishes as $\lambda_{EE}$, the on-the-job search probability, increases.

   — The model extension in Bloesch et al. (2025).

3. We examine several extensions of the model, including the effects of demand shocks (such as monetary policy shocks) and scenarios in which on-the-job search intensity depends on the real wage, following Pilossoph and Ryngaert (2024).

We consider 3 different points in time: $t = 0, 1, 2$. At $t = 0$, the economy is at its steady-state: the number of employed is $\bar{N}$, that of unemployed is $\bar{U} = 1 - \bar{N}$. For simplicity, we assume that at $t = 2$, the economy gets back to the steady state, regardless of what happens at the interim period, $t = 1$.

**Structure of the Economy** Following Bloesch et al. (2025), there are two goods: an endowment good $X_t$ and services $Y_t$. They are combined into an aggregate consumption good, $C_t$, according to:

$$C_t = \left( \alpha_Y^{\frac{1}{\eta}} Y_t^{\frac{\eta-1}{\eta}} + \alpha_X^{\frac{1}{\eta}} X_t^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \tag{1}$$

with corresponding aggregate price index[1]

$$P_t = \left( \alpha_Y P_{y,t}^{1-\eta} + \alpha_X P_{x,t}^{1-\eta} \right)^{\frac{1}{1-\eta}}. \tag{2}$$

[1] We assume $\alpha_X + \alpha_Y = 1$ with $\alpha_X > 0$ and $\alpha_Y > 0$ as usual.

Workers are hired by service firms to produce services $Y_t$, so that their real wage is determined by the nominal wage offered in that sector divided by the aggregate price level $P_t$. The total amount of endowment good $X_t$ is given, and $X_t$ is flexibly and competitively priced.

**Cost-of-Living Shock** Our cost of living shock is defined as a sudden decline in the endowment good $X_1$ in the interim period $t = 1$, which raises its price, $P_{x,1}$, and hence the price level $P_1$ from equation (2).

**Demand Block** Throughout this section, we assume that the policy rate is pegged, i.e., $i_t = \rho$ for $t = 0, 1$ so the households' Euler equation under log-preference, given by

$$\frac{1}{P_t C_t} = \frac{1 + i_t}{1 + \rho} \frac{1}{P_{t+1} C_{t+1}}$$

implies the intertemporal equalization of consumption expenditures, given by

$$P_0 C_0 = P_1 C_1 = P_2 C_2, \tag{3}$$

where $P_t$ is the price aggregator (of endowment good $X_t$ and service good $Y_t$ which is produced by firms) at time $t$, and $C_t$ is the corresponding consumption aggregator. Under the unit elasticity of substitution between $X_t$ and $Y_t$, i.e., $\eta = 1$ in the final aggregator (1), the households' expenditures on $X_t$ and $Y_t$ goods become proportional to each other, implying

$$\frac{P_{X,t} X_t}{\alpha_X} = \frac{P_{Y,t} Y_t}{\alpha_Y} = P_t C_t \tag{4}$$

for all $t = 0, 1, 2$. From (3) and (4), we obtain:

$$P_{Y,0} Y_0 = P_{Y,1} Y_1 = P_{Y,2} Y_2 \tag{5}$$

in equilibrium.

**Price Rigidity** For tractability, we temporarily assume perfect price rigidity in the service goods sector,[2] so that

$$P_{Y,0} = P_{Y,1} = P_{Y,2} = \bar{P}_Y,$$

which implies

$$Y_0 = Y_1 = Y_2 = \bar{Y}$$

where $\bar{Y}$ is the steady-state level of the service output. Therefore, service output $Y_1$ in the interim period $t = 1$ is at its steady-state level, $\bar{Y}$, 1 of the shocks realized at that time. This occurs because the economy is demand-driven, and households can perfectly smooth their consumption under a pegged monetary policy.

**Firm's Problem** We assume that perfectly-competitive retailers bundle service types $j$ according to a standard Dixit-Stiglitz production function with an associated ideal price index:

$$\begin{aligned} Y_t &= \left( \int \left(Y_t^j\right)^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{\epsilon}{\epsilon-1}}, \\ P_{y,t} &= \left( \int \left(P_{y,t}^j\right)^{1-\epsilon} dj \right)^{\frac{1}{1-\epsilon}}, \end{aligned}$$

yielding product demand for variety $j$:

$$\frac{Y_t^j}{Y_t} = \left( \frac{P_{y,t}^j}{P_{y,t}} \right)^{-\epsilon}. \tag{6}$$

Firm $i$, with its production function $Y_t^i = N_t^i$,[3] solves the following optimization at $t = 1$ facing nominal wage rigidities à la Rotemberg (1982), with its number of workers $N_0 = \bar{N}$ inherited from the previous period:

$$J(\bar{N}) = \max_{V_1^i, W_1^i} \bar{P}_Y N_1^i - W_1^i N_1^i - \underbrace{\kappa(W_1) \cdot V_1^i}_{\substack{\text{Vacancy} \\ \text{creation cost}}} - \underbrace{\frac{\psi^W}{2} \left( \frac{W_1^i}{\bar{W}} - 1 \right)^2 \bar{W} N_1^i}_{\text{Wage changing cost}} + \underbrace{\frac{1}{1+\rho} J(N_1^i)}_{\substack{\text{Next-period} \\ \text{value function}}} \tag{7}$$

[2] We will characterize the flexible price case later as a separate case.

[3] We abstract from labor productivity $A_t$ here. Note that with production function $Y_t^i = N_t^i$, we have $N_0 = N_1 = N_2 = \bar{N}$ from (5).

subject to

$$N_1^i = \bar{N} = (1 - S(W_1^i|W_1))\bar{N} + R(W_1^i|W_1)V_1^i, \tag{8}$$

where $\kappa(W_1)V_1^i$ is a vacancy-creation cost, where $\kappa(W_1)$ is a function of aggregate wage $W_1$. We will later mostly consider the case of $\kappa(W_1) = \kappa W_1$ (linear function). $S(W_1^i|W_1)$ and $R(W_1^i|W_1)$ are separation and retaining probabilities, respectively, that depend on the firm's individual wage $W_1^i$ and the aggregate wage $W_1$. Note that firm $i$ takes $N_1^i = \bar{N}$ as given, since the required level of labor is determined by the demand block described above.

For the functional forms of $S(W_1^i|W_1)$ and $R(W_1^i|W_1)$, we will later use the same functional form as in the original model of Bloesch, Lee and Weber (2025).

Solving (7) and (8) with $\mu_1^i$ as the Lagrange multiplier to (8) yields the followings:

- For vacancy $V_1^i$:

$$\mu_1^i = \frac{\kappa(W_1)}{R(W_1^i|W_1)} \tag{9}$$

  which implies: the value of each worker is equal to the expected cost of hiring the worker. The creation of one vacancy costs $\kappa(W_1)$ but each vacancy is filled with probability $R(W_1^i|W_1)$. This interpretation is provided in de la Barrera i Bardalet (2023) as well.

- Wage $W_1^i$:

$$\begin{aligned} N_1^i \left[1 + \psi^W \frac{W_1^i - \bar{W}}{\bar{W}}\right] &= \frac{\kappa(W_1)}{R(W_1^i|W_1)} \left[R'(W_1^i|W_1)V_1^i - S'(W_1^i|W_1)\bar{N}\right] \\ &= \frac{\kappa(W_1)}{R(W_1^i|W_1)} \left[\frac{R(W_1^i|W_1)}{W_1^i} \underbrace{\frac{R'(W_1^i|W_1)W_1^i}{R(W_1^i|W_1)}}_{=\varepsilon_{R,1}} V_1^i - \underbrace{\frac{S'(W_1^i|W_1)W_1^i}{S(W_1^i|W_1)}}_{=\varepsilon_{S,1}} \cdot \frac{S(W_1^i|W_1)}{W_1^i}\bar{N}\right] \end{aligned} \tag{10}$$

  which becomes

$$N_1^i \left[1 + \psi^W \frac{W_1^i - \bar{W}}{\bar{W}}\right] = \frac{\kappa(W_1)}{W_1^i} \left[\varepsilon_{R,1} \cdot V_1^i - \varepsilon_{S,1} \cdot \frac{S(W_1^i|W_1)}{R(W_1^i|W_1)}\bar{N}\right]. \tag{11}$$

- Envelope condition:

$$J'(\bar{N}) = (1 - S(W_1^i|W_1))\mu_1^i = (1 - S(W_1^i|W_1))\frac{\kappa(W_1)}{R(W_1^i|W_1)}. \tag{12}$$

Later, we will impose the (symmetric) equilibrium condition: $W_1^i = W_1$ and $N_1^i = N_1 = \bar{N}$.

**Search and Matching Process** As mentioned earlier, we assume that employed and unemployed workers share consumption risks such that $\frac{C_t^e}{C_t^u} = \xi$. Consequently, under the equilibrium condition where all firms make identical decisions (i.e., symmetric equilibrium)—i.e., $W_1^i = W_1$, $N_1^i = N_1$, $V_1^i = V_1$—the following definitions can be introduced:

- Labor market tightness $\theta_1$:[4]

$$\theta_1 = \frac{V_1}{\lambda_{EE}\bar{N} + 1 - \bar{N}} \tag{13}$$

  where $\lambda_{EE}$ is the on-the-job search intensity, and we use $N_0 = \bar{N}$.

- Retaining probability $R(W_1^i = W_1|W_1)$:

$$R(W_1|W_1) = g(\theta_1)\left(\frac{1}{2}\phi_{E,1} + \frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}\right) \tag{14}$$

  where $\phi_{E,1}$ and $\phi_{U,1} \equiv 1 - \phi_{E,1}$ are fractions of employed (i.e., on-the-job searchers) and unemployed among job seekers, given by

$$\phi_{E,1} = \frac{\lambda_{EE}\bar{N}}{\lambda_{EE}\bar{N} + 1 - \bar{N}}. \tag{15}$$

- Separation probability $S(W_1^i = W_1|W_1)$:

$$S(W_1|W_1) = \frac{1}{2}\lambda_{EE}f(\theta_1) + \frac{1}{1+\xi^\gamma}\lambda_{EU} \tag{16}$$

  where we assume zero automatic separation (i.e., $s = 0$ in our dynamic general equilibrium model), and $\lambda_{EU}$ is the exogenous job-quitting probability.

[4]Note that for any $t$,

$$\theta_t = \frac{V_t}{\lambda_{EE}N_{t-1} + 1 - N_{t-1}}.$$

- Elasticity $\varepsilon_{R,1}$ and $\varepsilon_{S,1}$: from (14) and (16), we obtain

$$\varepsilon_{R,1} = \gamma \cdot \left( \frac{\frac{1}{4}\phi_{E,1} + \phi_{U,1}\left(\frac{\xi^\gamma}{(1+\xi^\gamma)^2}\right)}{\frac{1}{2}\phi_{E,1} + \left(\frac{\xi^\gamma}{1+\xi^\gamma}\right)\phi_{U,1}} \right) \simeq \gamma \cdot \left( \frac{\frac{1}{4}\phi_{E,1}}{\frac{1}{2}\phi_{E,1} + \left(\frac{\xi^\gamma}{1+\xi^\gamma}\right)\phi_{U,1}} \right), \tag{17}$$

and

$$\varepsilon_{S,1} = -\gamma \cdot \left( \frac{f(\theta_1)\lambda_{EE}\frac{1}{4} + \lambda_{EU}\frac{\xi^r}{(1+\xi^r)^2}}{0.5\lambda_{EE}f(\theta_1) + \left(\frac{1}{1+\xi^\gamma}\right)\lambda_{EU}} \right) \simeq -\frac{\gamma}{2}. \tag{18}$$

where we approximate $\frac{\lambda_{EU}}{1+\xi^\gamma} \simeq 0$ and $\frac{\phi_{U,1}\xi^\gamma}{(1+\xi^\gamma)^2} \simeq 0$, which hold well under our calibration. In equation (17), this approximation reflects the idea that the effect of higher wages on inducing unemployed workers to accept jobs at a firm is small relative to the effect of attracting on-the-job searchers from other firms.

**Equilibrium Characterization** Since all firms choose the same decisions in symmetric equilibrium, i.e., $W_1^i = W_1$, $V_1^i = V_1$, and $N_1^i = N_1 = \bar{N}$, from (14) and (16), we obtain

$$\begin{aligned} \frac{S(W_1|W_1)\bar{N}}{R(W_1|W_1)} &= \frac{\frac{1}{2}\lambda_{EE}\underbrace{f(\theta_1)}_{=\theta_1 g(\theta_1)}\bar{N} + \frac{1}{1+\xi^\gamma}\lambda_{EU}\bar{N}}{g(\theta_1)\left(\frac{1}{2}\phi_{E,1} + \frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}\right)} \\ &= \frac{\frac{1}{2}\phi_{E,1}g(\theta_1)V_1 + \frac{1}{1+\xi^\gamma}\lambda_{EU}\bar{N}}{g(\theta_1)\left(\frac{1}{2}\phi_{E,1} + \frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}\right)}. \end{aligned} \tag{19}$$

We then plug in (17), (18), and (19) to (11) to obtain

$$\bar{N}\left[1 + \psi^W \frac{W_1 - \bar{W}}{\bar{W}}\right] = \frac{\kappa(W_1)}{W_1}\left\{ \underbrace{V_1\left[\gamma\left(\frac{\frac{1}{2}\phi_{E,1}}{\frac{1}{2}\phi_{E,1} + \frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}}\right)\right]}_{\equiv\varepsilon_{11}} + \underbrace{\frac{\frac{\gamma}{2}\frac{1}{1+\xi^\gamma}\lambda_{EU}\bar{N}}{\left(\frac{1}{2}\phi_{E,1} + \frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}\right)g(\theta_1)}}_{\equiv\varepsilon_{21}} \right\}, \tag{20}$$

where $\varepsilon_{11} + \varepsilon_{21}$ in (20) becomes the 'effective' labor supply elasticity each firm faces. $\varepsilon_{11}$ is about the elasticity due to those who are on-the-job search: an increase in wage attracts more on-the-job searchers from other firms and reduce the endogenous separation of current workers, and given

other variables, this effect becomes more pronounced with higher measure of on-the-job searchers among job seekers, i.e., higher $\phi_{E,1}$ (thereby decrease in $\phi_{U,1}$). Eventually in equilibrium, every firm sets the same wage: $W_1^i = W_1$ for $\forall i$.

$\varepsilon_{21}$ is the elasticity attributed to those who quit their jobs to be unemployed: a higher wage deters workers from going to be unemployed. Compared with the recruiting probability $R(W_1|W_1)$,[5] the proportion of those who exit the labor force becomes larger under a bigger and more competitive job market with higher $\lambda_{EE}$, i.e., higher $\lambda_{EE}$ raises $\varepsilon_{21}$ and $\varepsilon_{11}$.

From (8), (14), and (16), we obtain the labor dynamics as follows:

$$\begin{aligned}\bar{N} = N_1 &= \left[1 - \left(\frac{1}{2}\lambda_{EE} \underbrace{f(\theta_1)}_{=\theta_1 g(\theta_1)} + \frac{1}{1+\xi^\gamma}\lambda_{EU}\right)\right]\bar{N} + g(\theta_1)\left(\frac{1}{2}\phi_{E,1} + \frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}\right)V_1 \\ &= \bar{N} - \bar{N}\frac{1}{1+\xi^\gamma}\lambda_{EU} + g(\theta_1)V_1\frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1},\end{aligned} \tag{21}$$

which implies

$$\frac{\bar{N}\frac{1}{1+\xi^\gamma}\lambda_{EU}}{\lambda_{EE}\bar{N} + 1 - \bar{N}} = f(\theta_1)\frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}. \tag{22}$$

Equations (20) and (22) constitute our equilibrium, with the demand condition $N_1 = Y_1 = \bar{N}$. From these two equations, we can theoretically determine the equilibrium values of $W_1$ and $V_1$.

**Cost-of-Living Shock, Again** As we assume in Bloesch, Lee and Weber (2025), the endowment good $X_t$ drops from its steady state level $\bar{X}$ to $X_1 < \bar{X}$ at $t = 1$ in an unanticipated manner. From equations (20) and (22), a sudden drop in $X_1$ from $\bar{X}$ does not affect the equilibrium levels of $V_1$ and $W_1$, and from the household's Euler equation (5), $N_1 = \bar{N}$ remains the same.

According to equation (4), the only variable that changes is the price of the endowment good $X_t$. In particular $P_{X,1}$ increases so that $P_{X,1}X_1 = \bar{P_X}\bar{X}$. Proposition 1 summarizes this result.

**Proposition 1** *A cost-of-living shock—defined as a sudden decrease in the endowment good $X_1$ from $\bar{X}$—does not change the equilibrium outcomes in the labor market. Employment, wages, and vacancies remain at their steady-state levels, that is, $N_1 = \bar{N}$, $W_1 = \bar{W}$, and $V_1 = \bar{V}$. The price $P_{X,1}$ of the endowment good $X_1$ increases in such a way that total expenditure remains the same,*

[5] As $\lambda_{EE}$ increases, $R(W_1|W_1)$ decreases from (14).

*that is,* $P_{X,1}X_1 = \bar{P}_X\bar{X}$.

## 1.1 Flexible price case

The irrelevance result for cost-of-living shocks in Proposition 1 remains valid even when firms are allowed to set prices fully flexibly.

Since firms are in monopolistic competition, represented by Dixit-Stiglitz aggregator with elasticity of substitution $\epsilon$, their demand function at $t = 1$ is given by

$$Y_1^i = Y_1 \left(\frac{P_{Y,1}^i}{P_{Y,1}}\right)^{-\epsilon}. \tag{23}$$

Each firm $i$ solves instead the following problem:[6]

$$J(\bar{N}) = \max_{P_{Y,1}^i, N_1^i, V_1^i, W_1^i} P_{Y,1}^i N_1^i - W_1^i N_1^i - \kappa(W_1)\cdot V_1^i - \frac{\psi^W}{2}\left(\frac{W_1^i}{\bar{W}} - 1\right)^2 \bar{W} N_1^i + \frac{1}{1+\rho} J(N_1^i) \tag{24}$$

subject to (23) and

$$Y_1^i = N_1^i = (1 - S(W_1^i|W_1))\bar{N} + R(W_1^i|W_1)V_1^i. \tag{25}$$

The solution to (24), with $W_1^i = W_1$, will be given by

$$\begin{aligned} P_{Y,1}^i = P_{Y,1} &= \frac{\epsilon}{\epsilon - 1}\left(W_1 + \frac{\kappa(W_1)}{R(W_1|W_1)} - \frac{1}{1+\rho}J'(N_1^i)\right) \\ &= \frac{\epsilon}{\epsilon - 1}\left(W_1 + \frac{\kappa(W_1)}{R(W_1|W_1)} - \frac{1}{1+\rho}(1 - S(W_2|W_2))\frac{\kappa(W_2)}{R(W_2|W_2)}\right) \end{aligned} \tag{26}$$

where $W_2 = \bar{W}$ as the economy returns to its steady state at $t = 2$.

The term $\frac{\kappa(W_1)}{R(W_1|W_1)}$ in (26) represents the cost of hiring through an additional vacancy. If a firm hires at $t = 1$, it can reduce its hiring requirement at $t = 2$ by one worker. The last term

$$\frac{1}{1+\rho}(1 - S(W_2|W_2))\frac{\kappa(W_2)}{R(W_2|W_2)}$$

captures this reduction in future hiring costs.[7] The first-order condition for $W_1^i$ under symmetric

[6] Note that under flexible prices, firms choose $N_1^i$ freely.

[7] The decomposition of marginal costs in equation (26) is similarly given in de la Barrera i Bardalet (2023).

equilibrium ($N_1^i = N_1$) yields:

$$N_1\left[1+\psi^W\frac{W_1-\bar{W}}{\bar{W}}\right]=\frac{\kappa(W_1)}{W_1}\left\{\underbrace{V_1\left[\gamma\left(\frac{\frac{1}{2}\phi_{E,1}}{\frac{1}{2}\phi_{E,1}+\frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}}\right)\right]}_{\equiv\varepsilon_{11}}+\underbrace{\frac{\frac{\gamma}{2}\frac{1}{1+\xi^\gamma}\lambda_{EU}\bar{N}}{\left(\frac{1}{2}\phi_{E,1}+\frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}\right)g(\theta_1)}}_{\equiv\varepsilon_{21}}\right\}, \tag{27}$$

From the demand block equation (5), we obtain

$$\bar{P}_Y\bar{Y}=P_{Y,1}Y_1=P_{Y,1}N_1. \tag{28}$$

Merging equations (26), (27), and (28), we obtain

$$\begin{aligned}\bar{P}_Y\bar{Y}=&\frac{\epsilon}{\epsilon-1}\left[W_1+\frac{\kappa(W_1)}{R(W_1|W_1)}-\frac{1}{1+\rho}(1-S(W_2|W_2))\frac{\kappa(W_2)}{R(W_2|W_2)}\right]\\&\cdot\frac{\kappa(W_1)}{W_1}\left\{\underbrace{V_1\left[\gamma\left(\frac{\frac{1}{2}\phi_{E,1}}{\frac{1}{2}\phi_{E,1}+\frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}}\right)\right]}_{\equiv\varepsilon_{11}}+\underbrace{\frac{\frac{\gamma}{2}\frac{1}{1+\xi^\gamma}\lambda_{EU}\bar{N}}{\left(\frac{1}{2}\phi_{E,1}+\frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}\right)g(\theta_1)}}_{\equiv\varepsilon_{21}}\right\}\\&\cdot\left[1+\psi^W\frac{W_1-\bar{W}}{\bar{W}}\right]^{-1}.\end{aligned} \tag{29}$$

In addition, the labor market dynamics is given by

$$N_1=\bar{N}-\bar{N}\frac{1}{1+\xi^\gamma}\lambda_{EU}+g(\theta_1)V_1\frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}. \tag{30}$$

Therefore, equations (29) and (30) fully describe the equilibrium in this case. Since (29) and (30) do not depend on $X_1$ or $P_{X,1}$, a cost-of-living shock, i.e., reduction in $X_1$ from $\bar{X}$, does not affect the labor market equilibrium outcome as in the rigid price case.

**Corollary 1** *Even when firms can adjust prices fully flexibly, a cost-of-living shock—defined as a sudden decline in $X_1$ from $\bar{X}$—does not alter the equilibrium outcomes in the labor market, i.e., $N_1=\bar{N}$, $W_1=\bar{W}$, and $V_1=\bar{V}$. The price $P_{X,1}$ of the endowment good $X_1$ increases in such a way that total expenditure remains unchanged, that is, $P_{X,1}X_1=\bar{P}_X\bar{X}$.*

Note that this result is independent of nominal wage rigidities: even if $\psi^W = 0$ (i.e., prices and wages are fully flexible), a cost-of-living shock still does not affect the labor market equilibrium.

## 1.2 Quits rate and wage growth under demand shocks

In this section, we revert to the assumption of rigid prices and demonstrate analytically that a positive demand shock leads to increases in both the on-the-job switching rate $\frac{1}{2}\lambda_{EE} f(\theta_1)$[8] and wage growth. As $f(\cdot)$ is increasing, it is equivalent to a positive correlation between market tightness $\theta_1$ and wage growth under a demand shock.

We begin by defining a positive demand shock as one that increases $N_1$ from $\bar{N}$. For example, a reduction in the policy rate at $t = 1$ would trigger a consumption boom, prompting firms to raise their labor demand at that time. Firms take the higher labor deman $N_1$ as given. We begin with our equilibrium conditions. With $N_1 > \bar{N}$, we have

$$N_1 \left[ 1 + \psi^W \frac{W_1 - \bar{W}}{\bar{W}} \right] = \frac{\kappa(W_1)}{W_1} \left\{ \underbrace{V_1 \left[ \gamma \left( \frac{\frac{1}{2}\phi_{E,1}}{\frac{1}{2}\phi_{E,1} + \frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}} \right) \right]}_{\equiv \varepsilon_{11}} + \underbrace{\frac{\frac{\gamma}{2}\frac{1}{1+\xi^\gamma}\lambda_{EU}\bar{N}}{\left( \frac{1}{2}\phi_{E,1} + \frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1} \right) g(\theta_1)}}_{\equiv \varepsilon_{21}} \right\}, \tag{31}$$

and

$$N_1 = \bar{N} - \bar{N}\frac{1}{1+\xi^\gamma}\lambda_{EU} + g(\theta_1) V_1 \frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}. \tag{32}$$

To obtain a more precise result, we log-linearize (32) and derive[9]

$$0 < \check{N}_1 = \frac{1}{1+\xi^\gamma}\lambda_{EU} \left( \underbrace{\frac{g'(\bar{\theta}_1)\bar{\theta}_1}{g(\bar{\theta}_1)}}_{\equiv -\varepsilon_{g,\theta}} \check{\theta}_1 + \check{\theta}_1 \right) = \frac{1}{1+\xi^\gamma}\lambda_{EU} \left( 1 - \underbrace{\varepsilon_{g,\theta}}_{<1} \right) \check{\theta}_1, \tag{33}$$

which shows the relationship between employment deviations $\check{N}_1$ and labor market tightness $\check{\theta}_1$. Note that in equation (33), the firm's matching elasticity $\varepsilon_{g,\theta} \geqslant 0$ of $g(\theta_1)$, a condition that holds

[8] The quits rate includes workers who voluntarily leave to become unemployed, though this margin is small relative to on-the-job switching.

[9] We use $\check{\theta}_1 = \check{V}_1$ because $\theta_1$ and $V_1$ are proportional and $\lambda_{EE}\bar{N} + 1 - \bar{N}$ is constant.

under various specifications.[10] Therefore, (33) implies that a positive demand shock tightens the labor market, so that $\check{\theta}_1 > 0$ whenever $\check{N}_1 > 0$.

For additional tractability, we now assume a linear form for the vacancy cost, $\kappa(W_1) = \kappa W_1$,[11] which leads to

$$N_1\left(1+\psi^W\frac{W_1-\bar{W}}{\bar{W}}\right) = \kappa\left\{\underbrace{V_1\left[\gamma\left(\frac{\frac{1}{2}\phi_{E,1}}{\frac{1}{2}\phi_{E,1}+\frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}}\right)\right]}_{\equiv\varepsilon_{11}} + \underbrace{\frac{\frac{\gamma}{2}\left(\frac{1}{1+\xi^\gamma}\right)\lambda_{EU}\bar{N}}{\left(\frac{1}{2}\phi_{E,1}+\frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}\right)g(\theta_1)}}_{\equiv\varepsilon_{21}}\right\}, \tag{34}$$

We log-linearize (34) and merge with (33) to obtain

$$\frac{1}{1+\xi^\gamma}\lambda_{EU}\left(1-\underbrace{\varepsilon_{g,\theta}}_{<1}\right)\check{\theta}_1+\psi^W\check{W}_1 = \left[\frac{\bar{\varepsilon}_{11}}{\bar{\varepsilon}_{11}+\bar{\varepsilon}_{21}}+\frac{\bar{\varepsilon}_{11}}{\bar{\varepsilon}_{11}+\bar{\varepsilon}_{21}}\varepsilon_{g,\theta}\right]\check{\theta}_1. \tag{35}$$

Since $\frac{1}{1+\xi^\gamma}\lambda_{EU}$ is small under our calibration, equation (33) implies that $\check{\theta}_1 > 0$ leads to $\check{W}_1 > 0$ in (35) as well. Consequently, wage changes and labor market tightness (on-the-job switching rate) move positively together, as summarized in Proposition 2.[12]

**Proposition 2** *When $\kappa(W_1) = \kappa W_1$, both the market tightness $\theta_1$ (and the corresponding on-the-job switching rate $0.5\lambda_{EE}f(\theta_1)$) and the wage $W_1$ rise in response to a positive demand shock.*

Thus, our simple model generates the benchmark results in Bloesch, Lee and Weber (2025).

## 1.3 With real benefits of unemployment

In this section, we assume that unemployed workers receive an inflation-indexed amount of consumption $b_1$ at $t = 1$. Under this assumption, all the previously stated equilibrium conditions—namely (13), (14), (15), (16), (17), (18), (20), and (22)—continue to hold, with

$$c(P_1, W_1) \equiv \frac{\left(\frac{W_1}{P_1}\right)^\gamma}{b_1^\gamma + \left(\frac{W_1}{P_1}\right)^\gamma}.$$

[10] Since $f(\theta_1) = \theta_1 g(\theta_1)$, $\varepsilon_{f,\theta} \equiv \frac{g'(\bar{\theta}_1)\bar{\theta}_1}{g(\bar{\theta}_1)} = 1 - \varepsilon_{g,\theta} > 0$ under our specification, as $f(\theta_1)$ is increasing in $\theta_1$.

[11] This corresponds to $\chi = 0$ in Bloesch, Lee and Weber (2025).

[12] Note that without wage nominal rigidities, i.e., $\psi^W = 0$, equilibrium might not exist.

in the position of $\frac{\xi^\gamma}{1+\xi^\gamma}$. Here, $b_1$ represents the consumption-equivalent benefit received during unemployment. Unemployed individuals compare this benefit with the real wage, $\frac{W_1}{P_1}$, when deciding whether to return to work.[13] We again examine the case of a cost-of-living shock, specifically a negative shock to $X_1$.

Note that $c(P_1, W_1)$ increases in $W_1$ and decreases in $P_1$, where $P_1$ is the total price aggregator of endowment good $X_1$ and service good $Y_1$. Under rigid service prices, i.e., $P_{Y,1} = \bar{P}_Y$, a cost-of-living shock as described above increases $P_1$ and lower $c(P_1, W_1)$. We ask how the economy's responses to a cost-of-living shock under this specification would differ from the above case where $c(P_1, W_1) \equiv \frac{\xi^\gamma}{1+\xi^\gamma}$. Intuitively, a rise in cost-of-living reduces the relative attractiveness of working compared with being unemployed, resulting in a lower $c(P_1, W_1)$, which leads to an increase in $W_1$.

As before, we assume $\kappa(W_1) = \kappa W_1$ for tractability. The equilibrium can be represented by

$$\frac{\bar{N}\left(1 - c(P_1, W_1)\right)\lambda_{EU}}{\lambda_{EE}\bar{N} + 1 - \bar{N}} = f(\theta_1)c(P_1, W_1)\phi_{U,1}. \tag{36}$$

and

$$\bar{N}\left[1 + \psi^W \frac{W_1 - \bar{W}}{\bar{W}}\right] = \kappa \left\{ \underbrace{V_1\left[\gamma\left(\frac{\frac{1}{2}\phi_{E,1}}{\frac{1}{2}\phi_{E,1} + c(P_1, W_1)\phi_{U,1}}\right)\right]}_{\equiv \varepsilon_{11}} + \underbrace{\frac{\frac{\gamma}{2}\left(1 - c(P_1, W_1)\right)\lambda_{EU}\bar{N}}{\left(\frac{1}{2}\phi_{E,1} + c(P_1, W_1)\phi_{U,1}\right) g(\theta_1)}}_{\equiv \varepsilon_{21}} \right\}, \tag{37}$$

where we rely on the fact that output (and labor) remains at the steady state level due to households' perfect consumption smoothing. We assume that at the steady state, $c(\bar{P}_1, \bar{W}_1) = \bar{c} = \frac{\xi^\gamma}{1+\xi^\gamma}$. First, we log-linearize (36) and obtain

$$-\frac{\bar{c}}{1-\bar{c}}\check{c} = \underbrace{\frac{f'(\bar{\theta}_1)\bar{\theta}_1}{f(\bar{\theta}_1)}}_{\equiv \varepsilon_{f,\theta}} \check{\theta}_1 + \check{c} \tag{38}$$

with

$$\check{c} = \frac{\bar{c}_P \bar{P}_1}{\bar{c}}\check{P}_1 + \frac{\bar{c}_W \bar{W}_1}{\bar{c}}\check{W}_1. \tag{39}$$

[13] We assume that the monetary authority sticks to pegging its policy rate.

Equations (38) and (39) yield

$$\check{\theta}_1 = -\frac{1}{(1-\bar{c})\varepsilon_{f,\theta}}\left(\frac{\bar{c}_P\bar{P}_1}{\bar{c}}\check{P}_1 + \frac{\bar{c}_W\bar{W}_1}{\bar{c}}\check{W}_1\right). \tag{40}$$

By log-linearizing (37), we obtain

$$\psi^W\check{W}_1 = \frac{\bar{\varepsilon}_{11}}{\bar{\varepsilon}_{11}+\bar{\varepsilon}_{21}}\left[\check{\theta}_1 - \frac{\bar{c}\phi_{U,1}}{\frac{1}{2}\phi_{E,1}+\bar{c}\phi_{U,1}}\check{c}\right] + \frac{\bar{\varepsilon}_{21}}{\bar{\varepsilon}_{11}+\bar{\varepsilon}_{21}}\left[-\frac{\bar{c}}{1-\bar{c}}\check{c} - \frac{\bar{c}\phi_{U,1}}{\frac{1}{2}\phi_{E,1}+\bar{c}\phi_{U,1}}\check{c} \underbrace{-\frac{g'(\bar{\theta}_1)\bar{\theta}_1}{g(\bar{\theta}_1)}}_{\equiv\varepsilon_{g,\theta}}\check{\theta}_1\right]. \tag{41}$$

With (40) and (41), in equilibrium, the equilibrium wage $\check{W}_1$ is given by

$$\check{W}_1 = \frac{d_W}{\psi^W\frac{1}{\gamma(1-\bar{c})} + d_W}\check{P}_1 < \check{P}_1, \tag{42}$$

with labor market tightness $\check{\theta}_1$ given by

$$\check{\theta}_1 = -\frac{\gamma}{\varepsilon_{f,\theta}}\left(\check{W}_1 - \check{P}_1\right) > 0.$$

Equation (42) shows that wages $W_1$ increase in response to a cost-of-living shock, but at a slower rate than the price aggregator ($\check{W}_1 < \check{P}_1$), resulting in a decline in the real wage. Because the cost-of-living shock makes work less attractive, firms face greater difficulty in hiring, causing labor market tightness (and vacancies) to rise.[14] This result is summarized in the following proposition.

**Proposition 3** *When $\kappa(W_1) = \kappa W_1$, the wage $W_1$ increases in response to a cost-of-living shock, but it rises at a slower rate than the price aggregator, meaning $\check{W}_1 < \check{P}_1$. As a result, the labor market becomes tighter, which is reflected in $\check{\theta}_1 > 0$.*

**Endogenous Real Wage Rigidity** Notice in (42), when $\psi^W = 0$—corresponding to fully flexible wages—we have $\check{W}_1 = \check{P}_1$, so the real wage becomes endogenously rigid. In this case, all other labor market variables, including vacancies, tightness, and employment, also remain unchanged in response to a cost-of-living shock.

[14]This is because firms must produce service output based on aggregate demand, which they take as given when service prices are rigid.

## 1.4 Variable On-the-Job Search Intensity

Following Pilossoph and Ryngaert (2024), in this section, we assume that the on-the-job probability $\lambda_{EE}$ at $t = 1$ is given by

$$\lambda_{EE}(P_1, W_1) \equiv \bar{\lambda}_{EE} \left(\frac{\bar{W}_1}{\bar{P}_1}\right)^m \left(\frac{W_1}{P_1}\right)^{-m} \tag{43}$$

with $m = 4$. Equation (43) assumes that workers increase their on-the-job search when the real wage declines. In this case, a cost-of-living shock raises $\lambda_{EE,1}$, giving firms an incentive to increase wages in order to retain their workers. Note from

$$\phi_{E,1} = \frac{\lambda_{EE}\bar{N}}{\lambda_{EE}\bar{N} + 1 - \bar{N}}, \quad \phi_{U,1} = \frac{1 - \bar{N}}{\lambda_{EE}\bar{N} + 1 - \bar{N}}, \quad \theta_1 = \frac{V_1}{\lambda_{EE}\bar{N} + 1 - \bar{N}}, \tag{44}$$

that a higher $\lambda_{EE}$ increases $\phi_{E,1}$ and decreases $\phi_{U,1}$, meaning that a larger share of job seekers are engaged in on-the-job search. With $\kappa(W_1) = \kappa W_1$, the equilibrium is characterized by:

$$\begin{aligned} & N_1 \left(1 + \psi^W \frac{W_1 - \bar{W}}{\bar{W}}\right) \\ & = \kappa \left\{ \underbrace{\underbrace{\left(\lambda_{EE}(P_1, W_1)\bar{N} + 1 - \bar{N}\right) \theta_1}_{=V_1} \left[\gamma \left(\frac{\frac{1}{2}\phi_{E,1}}{\frac{1}{2}\phi_{E,1} + \frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}}\right)\right]}_{\equiv \varepsilon_{11}} + \underbrace{\frac{\frac{\gamma}{2}\frac{1}{1+\xi^\gamma}\lambda_{EU}\bar{N}}{\left(\frac{1}{2}\phi_{E,1} + \frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1}\right) g(\theta_1)}}_{\equiv \varepsilon_{21}} \right\}, \end{aligned} \tag{45}$$

and

$$\begin{aligned} N_1 &= \bar{N} - \bar{N}\frac{1}{1+\xi^\gamma}\lambda_{EU} + g(\theta_1)V_1\frac{\xi^\gamma}{1+\xi^\gamma}\phi_{U,1} \\ &= \bar{N} - \bar{N}\frac{1}{1+\xi^\gamma}\lambda_{EU} + f(\theta_1)\frac{\xi^\gamma}{1+\xi^\gamma}\left(1 - \bar{N}\right). \end{aligned} \tag{46}$$

**Sticky Prices** Unlike Sections 1.2 and 1.3, where we assume fully rigid prices, here we consider a flexible form of price stickiness. In this case, instead of only an increase in $W_1$, the service price $P_{Y,1}$ also rises to some extent. Specifically, we assume that $\check{P}_{Y,1} = d_P\check{W}_1$, with $d_P > 0$, where $\check{P}_{Y,1}$ and $\check{W}_1$ represent log-deviations from their respective steady-state levels. The case $d_P = 0$ corresponds to perfectly rigid prices.

Since $P_{Y,1}N_1 = \bar{P}_Y\bar{Y}$ holds—because households maintain equal expenditure under a pegged monetary policy—we know

$$\check{N}_1 = -\check{P}_{Y,1} = -d_P\check{W}_1 = \frac{1}{1+\xi^\gamma}\lambda_{EU}\underbrace{\varepsilon_{f,\theta}}_{>0}\check{\theta}_1 \tag{47}$$

where the last equality is from (46). From (47), it follows that if $\check{W}_1 > 0$ in equilibrium in response to a cost-of-living shock (i.e., $\check{P}_1 > 0$), then $\check{\theta}_1 < 0$, meaning that the labor market becomes less tight. With a lower $\theta_1$, the wage $\check{W}_1$ increases less in response to $\check{P}_1 > 0$ in (45), because $\theta_1$ enters $\varepsilon_{11}$ and $g(\theta_1)$ is decreasing in $\theta_1$. In a less tight labor market, firms do not need to raise wages as much to attract job seekers and retain potential leavers.

By log-linearizing (44), we obtain

$$\check{\phi}_{E,1} = \bar{\phi}_{U,1}\check{\lambda}_{EE}, \quad \check{\phi}_{U,1} = -\bar{\phi}_{E,1}\check{\lambda}_{EE} \tag{48}$$

with $\check{\lambda}_{EE} = -m\left(\check{W}_1 - \check{P}_1\right)$. Linearizing (45) yields:

$$\begin{aligned}\check{N}_1 + \psi^W\check{W}_1 =& \frac{\bar{\varepsilon}_{11}}{\bar{\varepsilon}_{11}+\bar{\varepsilon}_{21}}\left[\bar{\phi}_{E,1}\check{\lambda}_{EE} + \check{\theta}_1 + (1-\chi)\check{\lambda}_{EE}\right] \\ &- \frac{\bar{\varepsilon}_{21}}{\bar{\varepsilon}_{11}+\bar{\varepsilon}_{21}}\left[\chi\check{\phi}_{E,1} + (1-\chi)\check{\phi}_{U,1} - \varepsilon_{g,\theta}\check{\theta}_1\right],\end{aligned} \tag{49}$$

where

$$\chi \equiv \frac{\frac{1}{2}\bar{\phi}_{E,1}}{\frac{1}{2}\bar{\phi}_{E,1} + \frac{\xi^\gamma}{1+\xi^\gamma}\bar{\phi}_{U,1}}.$$

By combining (47), (48), and (49) with $\check{\lambda}_{EE} = -m(\check{W}_1 - \check{P}_1)$ and approximating $\frac{\bar{\varepsilon}_{21}}{\bar{\varepsilon}_{11}+\bar{\varepsilon}_{21}} \simeq 0$ with $\frac{\bar{\varepsilon}_{11}}{\bar{\varepsilon}_{11}+\bar{\varepsilon}_{21}} \simeq 1$ in (49) as before, we obtain

$$\check{W}_1 = \frac{m\left(\bar{\phi}_{EE} + 1 - \chi\right)}{\psi^W - d_P + m\left(\bar{\phi}_{EE} + 1 - \chi\right) + \frac{d_P}{\frac{\lambda_{EU}}{1+\xi^\gamma}\varepsilon_{f,\theta}}}\check{P}_1 > 0. \tag{50}$$

**Interpretation** Under fully rigid prices ($d_P = 0$), we have

$$\check{W}_1 = \frac{m\left(\bar{\phi}_{EE} + 1 - \chi\right)}{\psi^W + m\left(\bar{\phi}_{EE} + 1 - \chi\right)}\check{P}_1 > 0. \tag{51}$$

with $\check{\theta}_1 = 0$, indicating no change in labor market tightness. When employees intensify their on-the-job search, firms respond by raising wages and posting more vacancies so that market tightness remains the same.[15] As a result, other labor market variables also remain unchanged. This occurs because, under fully rigid prices, labor demand does not adjust in response to a negative shock to $X_1$. Note again that when wages are fully flexible ($\psi^W = 0$), we have $\check{W}_1 = \check{P}_1$, implying that the real wage becomes endogenously rigid.

Under sticky prices, according to (47), we have $\check{\theta}_1 < 0$ and $\check{W}_1 > 0$, as shown in (50), consistent with the findings of Bloesch, Lee and Weber (2025). In equilibrium, firms increase the service price in response to a cost-of-living shock, which reduces both service and labor demand. At the same time, a higher probability of on-the-job search among workers lowers labor market tightness $\theta_1$. This, in turn, decreases firms' incentive to raise wages to attract job seekers, resulting in a muted wage response. This effect is captured by the term $\frac{d_P}{\frac{\lambda_{EU}}{1+\xi^\gamma}\varepsilon_{f,\theta}}$ in the denominator of (50).

On the other hand, when service firms face lower labor demand, the marginal cost of raising wages—in terms of total wage bills (i.e., the dollar increase applies to all workers, both new hires and incumbents)—is smaller from each firm's perspective. This increases firms' incentive to raise wages, an effect captured by the $d_P$ term in (50). In effect, the first effect dominates the second effect,[16] and wage increase under endogenous on-the-job search intensity is muted following (43).

Finally, even when $d_P = 0$, we see that $\check{W}_1 < \check{P}_1$ under $m > 0$, where $\check{W}_1$ is increasing in $m$, which confirms our intuition in Bloesch, Lee and Weber (2025) that rising separations and wage growth push firms to want to raise wages in response. Note that with $m = 0$, we obtain $\check{W}_1 = 0$, consistent with our baseline result in Proposition 1.

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[15] Note that with $\psi^W > 0$, $\check{W}_1 < \check{P}_1$ in equation (51), which implies that $\check{\lambda}_{EE} > 0$. Therefore, in order for $\check{\theta}_1$ to be constant, we must have $\check{V}_1 > 0$.

[16] Remember $\frac{\lambda_{EU}}{1+\xi^\gamma}$ is small.

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